

B.Sc. (Hons.) SEMESTER III EXAMINATION 2022-23
STATISTICS

Paper No. STB-301: Statistical Inference

Time: Three hours

Max. Marks: 70

NOTE: Attempt any five questions. All questions carry equal marks.

1. (a) What do you mean by a random sample from a distribution? Distinguish between standard deviation and standard error. Show that the sampling distribution of the sum of two independent Poisson variates is again a Poisson variate.
(b) What are the criteria of a good estimator? Explain any two of them with examples. Let $X_1, X_2, \dots, X_n$ be a random sample from an $N(\mu, \sigma^2)$ population and $T = \frac{1}{n} \sum_{i=1}^n X_i$ be the statistic based on sample values. Show that (i) $E(T) = \mu$, (ii) $V(T) = \frac{\sigma^2}{n}$.
2. Define null and alternative hypothesis with a suitable example. Explain the concept of Type-I and Type-II errors and bring out their importance in testing of hypothesis problems. A random sample is taken from a Poisson distribution with parameter λ . If $x > 2$ is the critical region for testing the hypothesis $H_0 : \lambda = 2$ against $H_1 : \lambda = 1$, find the size and power of the test.
3. (a) Define chi-square statistic. Describe the chi-square test for testing the independence of attributes.
(b) What is F -distribution? Explain the testing procedure for testing the equality of two variances from two independent normal populations.
4. State the test of significance for differences of two means from two independent normal populations when variances are equal but unknown. Below are given the gain in weights (in kgs) of pigs fed on two diets A and B:

Diet A: 32, 35, 33, 29, 24, 41, 42, 36

Diet B: 35, 36, 39, 37, 38, 25, 26, 30, 24, 20

Test if the two diets differ significantly as regards their effect on increase in weight at a 5% level of significance, given that $t_{16,0.05} = 2.12$.

5. (a) Describe the importance of the central limit theorem. State the test of significance for differences of two proportions in the case of a large sample.
(b) Describe the chi-square goodness of fit test by stating its required assumptions.
6. Define order statistics with its different applications. Show that the distribution of the r^{th} order statistic for a uniform distribution $U(0, 1)$ follows a beta distribution. Also deduce the distribution of the smallest and largest order statistics.
7. (a) Explain the advantages and disadvantages of non-parametric tests. Describe the one-sample Kolmogorov-Smirnov non-parametric test.
(b) Develop the median non-parametric test by stating the underlying assumptions and the null hypothesis.
8. Write short notes on any **three** of the following:
 - (a) Estimate and estimator
 - (b) Sampling distribution

- (c) Simple and composite hypothesis with examples
 - (d) Fisher's t -distribution and its uses
 - (e) Fisher's z -transformation
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External Portals & Academic Resources

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app/>